

Simultaneous complement avoidance and explicit Thue–Morse barriers

20 September 2026 — continuation for discussion

This continuation resolves two questions left in the earlier packet: simultaneous avoidance of the two complementary Thue–Morse sequences, and an explicit finite matching bound for the particular Thue–Morse population. The first result holds for every non-eventually-periodic binary sequence. These are hand proofs with finite computational controls; independent review and priority checking remain. The earlier Lean files do not certify these additions.

1. One population avoids both complementary targets

For a binary sequence s , write $\bar{s}(k) = 1 - s(k)$ and put

$$r(2j) = s(j), \quad r(2j+1) = 1 - s(j).$$

The population P_s has vertices $\mathbb{N}$, birthdate v at vertex v , and, for every $v \geq 2$, precisely the incoming edges

$$v-1 \longrightarrow v \text{ labelled } r(v), \quad v-2 \longrightarrow v \text{ labelled } 1 - r(v).$$

There is no edge $0 \rightarrow 1$. The roots are 0 and 1; every non-root has a parent of each label; every vertex has at most two children. All birthdate sublevel sets are finite. Thus this is an Alexander population in the edge-labelled convention, with no restriction on a realizing path’s start.

Theorem. If s is not eventually periodic, P_s realizes neither s nor $\bar{s}$.

The original packet proves avoidance of s . Here is a self-contained joint argument. Let $v_0, v_1, \dots$ be a hypothetical matching path and put $d_k = v_k - 2k$. Every increment of d_k is -1 or 0 .

For target s , induction gives $v_k \geq 2k$: at $v_k = 2k$, the increment-one edge, if present, has label $r(2k+1) = 1 - s(k)$ and cannot match. Hence $d_k \geq 0$ and eventually $d_k = d$ is constant.

For target $\bar{s}$, the corresponding induction gives $v_k \geq 2k - 1$. If $v_k > 2k - 1$, any edge preserves the next lower bound. At equality, an increment-one edge has label $r(2k) = s(k)$ and cannot match. Thus $d_k \geq -1$ and again it stabilizes. In fact an infinite matching path cannot visit $d_k = -1$: both outgoing edges there have label $s(k)$, since the increment-two label is $1 - r(2k+1) = s(k)$. This also handles that exceptional offset without invoking a shift of length zero.

On the constant-offset tail, every increment is two. For $d \geq 0$, put $q = \lfloor d/2 \rfloor + 1 > 0$. The matching equation is as follows:

For target s : even d gives $s(k) = 1 - s(k + q)$, and odd d gives $s(k) = s(k + q)$. For target $\bar{s}$: even d gives $s(k) = s(k + q)$, and odd d gives $s(k) = 1 - s(k + q)$.

Either equality makes s eventually periodic, with period q or $2q$. This proves the theorem. The boundary -1 , rather than 0 , is the only new ingredient needed for the complementary target.

For a periodic control of period p , the construction correctly fails to avoid either target: the increment-two paths starting at $2p - 1$ and $2p - 2$ realize s and $\bar{s}$, respectively. These controls include $p = 1$ and the actual root convention.

Consequence for the two-phase substitution gadget. The earlier exact relation is

$$\mu(x) \in L(Q(P)) \iff x \in L(P) \text{ or } \bar{x} \in L(P), \quad \mu(0) = 01, \mu(1) = 10.$$

Therefore $Q(P_s)$ avoids $\mu(s)$ and $\mu(\bar{s})$ whenever s is aperiodic. For Thue–Morse t , both t and $\bar{t}$ are fixed by μ , so P_t and every iterate $Q^n(P_t)$ avoid both. This settles the simultaneous-avoidance question that previously prevented this particular use of the gadget. It does not construct an intersection operation for arbitrary population languages.

2. A short-run lemma for Thue–Morse

Write $t(k) = \text{popcount}(k) \bmod 2$, and, for $q \geq 1$, define

$$c_q(k) = t(k) \text{ xor } t(k + q), \quad \lambda(q) = 2^{\nu_2(q)}.$$

Lemma. Every constant consecutive run of c_q has length at most $3\lambda(q)$, independently of where the run starts.

The identities $t(2n) = t(n)$ and $t(2n + 1) = 1 - t(n)$ imply that t has no three consecutive equal bits: any three positions include an even/odd pair $(2n, 2n + 1)$.

First let $q = 2r + 1$ be odd. Then

$$\begin{aligned} c_q(2n) &= 1 \text{ xor } t(n) \text{ xor } t(n + r), \\ c_q(2n + 1) &= 1 \text{ xor } t(n) \text{ xor } t(n + r + 1). \end{aligned}$$

Four equal values starting at $2n$ would force $t(n + r) = t(n + r + 1) = t(n + r + 2)$, by comparing the two aligned pairs. Four equal values starting at $2n + 1$ would instead force $t(n) = t(n + 1) = t(n + 2)$, by comparing the pairs $(2n + 1, 2n + 2)$ and $(2n + 3, 2n + 4)$. Both are impossible. Thus the bound is three for odd q .

Finally $c_{2q}(2n) = c_{2q}(2n + 1) = c_q(n)$. Each doubling of the shift doubles each constant run. Removing all powers of two proves the lemma. This elementary symbolic-sequence lemma is used here as a tool; no claim that the lemma itself is new is intended.

3. An explicit bound for every starting vertex

For the Thue–Morse population P_t , $r = t$. Define

$$W(i) = \sum_{d=0}^i \lambda(\lfloor d/2 \rfloor + 1).$$

Theorem. A path starting at vertex i and matching a prefix of t has at most $i + 3W(i)$ edges. A path matching a prefix of $\bar{t}$ has at most $i + 1 + 3W(i)$ edges. In particular neither target can match

$$B(i) = i + 2 + 3W(i)$$

edges from that vertex. This is a sufficient barrier, not a claimed optimal one.

To prove it, separate increment-one edges from maximal consecutive runs of increment-two edges. Each increment-one edge decreases d_k by one; there are at most i such edges for target t and at most $i + 1$ for target $\bar{t}$. Each nonnegative offset d is visited in at most one plateau. On its increment-two run, the equations above make $c_q(k)$ constant, where $q = \lfloor d/2 \rfloor + 1$. The lemma bounds this run by $3\lambda(q)$. Summing over $0 \leq d \leq i$ proves the stated bounds. Offset -1 contributes no edges for the complementary target.

The bound grows as $O((i + 1) \log(i + 2))$. For an explicit estimate put $m = \lfloor i/2 \rfloor + 1$ and

$$S(m) = \sum_{q=1}^m \lambda(q) = m + \sum_{j \geq 1} 2^{j-1} \lfloor m/2^j \rfloor \leq m \left(1 + \frac{1}{2} \lfloor \log_2 m \rfloor\right).$$

Since $W(i) \leq 2S(m)$,

$$B(i) \leq i + 2 + 6m + 3m \lfloor \log_2 m \rfloor.$$

For exact integer evaluation, $S(0) = 0$ and $S(m) = \lceil m/2 \rceil + 2S(\lfloor m/2 \rfloor)$. Also $W(2h+1) = 2S(h+1)$ and $W(2h) = 2S(h) + \lambda(h+1)$.

This replaces a qualitative termination argument for a *fixed target* with a computable a priori cutoff. It does not contradict the earlier absence of a target-uniform bound at fixed start: an arbitrary aperiodic target may begin with an arbitrarily long constant block.

4. Checks, limitations and remaining questions

The accompanying `evidence/44s-complement-barrier-checks.py` enumerates finite paths directly from the graph’s two incoming-edge formulas, checks both boundary invariants on all short binary targets, uses periodic positive controls, and compares actual Thue–Morse first-empty times with $B(i)$. It also checks the short-run lemma over finite ranges. The JSON records the precise ranges and outcomes; finite checks are not the infinite proofs.

The logarithmic factor may be avoidable: the proof separately charges every offset’s longest possible run even when those runs cannot be concatenated along one matching path. A sharp asymptotic or

exact barrier remains open locally. **September 21 continuation:** `countable-avoidance.md` now settles simultaneous avoidance for arbitrary countable aperiodic families, including finite collections, at optimal degree for every finite alphabet. Other embedding/homomorphism versions of universal avoiding populations and more informative ordinal invariants remain outside these results.

Source: Samuel A. Alexander, *Biologically unavoidable sequences*, Electronic Journal of Combinatorics **20**(1) (2013), P31, doi:10.37236/3035. The population and earlier classification proof are in this packet's `results.md`. The claims here are local extensions of that construction, not attributed to Alexander.